

Title

Algorithmic Fairness Beyond Technical Metrics: A Focused Literature Review

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Introduction

Algorithmic decision-making systems are commonly used in areas that affect individuals' lives, such as hiring, criminal justice, and access to opportunities. As these systems become more widespread and influential, relying solely on predictive performance raises important concerns about fairness. However, there is no single definition of fairness. Fairness can be evaluated according to different criteria, each reflecting a different understanding of what constitutes a fair outcome. It is therefore insufficient to ask whether a model is fair without specifying the standards according to which fairness is being assessed. This review examines how algorithmic fairness should be conceptualized when mathematical definitions reflect different assumptions and may fail to capture broader sources of bias.

Competing Definitions of Fairness

One of the main challenges in evaluating fairness is that it can be represented by a variety of mathematical criteria. To better understand this challenge, it is useful to consider two key criteria: demographic parity and equal opportunity. Demographic parity focuses on whether different groups receive positive predictions at equal rates. Equal opportunity, on the other hand, focuses on whether individuals who actually belong to the positive class have an equal chance of receiving a positive prediction across groups (Mitchell et al., 2021). For example, if two groups have different proportions of qualified applicants, a model could select the same proportion of qualified individuals from each group, thereby satisfying equal opportunity. However, the overall selection rates of the two groups could still differ, meaning that demographic parity would not be satisfied (Barocas et al., 2023; Mitchell et al., 2021). Hence, fulfilling one fairness criterion does not necessarily mean it will satisfy another. Equalized odds extends the idea of equal opportunity by considering both true-positive and false-positive rates across groups (Mitchell et al., 2021). This adds another way of looking at fairness, since it considers not only who receives a positive outcome, but also who is more likely to be affected by the model's mistakes.

Looking at these different definitions makes it difficult to say that there is one clear way to measure fairness. A model can seem fair under one definition but unfair under another. Because of this, choosing a fairness measure is also about deciding what we actually want to be equal between different groups. This shows that fairness cannot be decided by numbers alone, since we first have to decide what we consider fair in that situation.

Sources of Bias in Machine Learning

Even if we all agreed on a definition of fairness, there would still be another problem to tackle: bias can enter a machine-learning system in several different ways. One example worth mentioning is historical bias, which occurs when the data used to train a model already reflects inequalities that exist in society (Mehrabi et al., 2021). For instance, if a hiring model is trained on a company's previous hiring decisions, it may learn patterns from decisions that were already influenced by existing biases. This means that the model could accurately learn from its data while still reproducing unfair patterns. Representation bias, on the other hand, can occur when particular groups are underrepresented in the training data. As a result, the model may perform well for the majority population while performing less accurately for groups that are not sufficiently represented (Mehrabi et al., 2021). Bias can appear from the way we measure something. Sometimes we can't obtain the outcome we want directly, therefore we need another variable as a substitute. For example, a hiring system may use past performance ratings as a measure of employee quality. However, if these ratings are subjective or already influenced by bias, the model may learn from a problematic measure of what it is supposed to predict. All of these examples show that unfairness can exist before an algorithm even begins making predictions. Therefore, making the model itself "fair" may not solve the entire problem if the data it learns from already contains bias.

Fairness Beyond the Algorithm

Even if bias in the data is identified and an appropriate fairness measure is chosen, evaluating the algorithm alone may still give us an incomplete picture of fairness. Selbst et al. (2019) argue that algorithmic systems should not be viewed as isolated technical tools, but as parts of larger sociotechnical systems. For instance, when evaluating a hiring algorithm, focusing only on the relationship between an applicant's data and the model's final prediction leaves out other parts of the hiring process. However, the hiring process includes much more. The opportunities available to an applicant before applying, the criteria used by the company, the decisions made by recruiters, and the final interview process can all influence the outcome. This means that even if the algorithm satisfies a specific fairness measure, the overall hiring process may still contain inequalities. Selbst et al. (2019) describe this problem through the Framing Trap, which occurs when the boundaries of an algorithmic system are defined too narrowly and important social factors are left outside the analysis.

Another relevant concept is the Formalism Trap. It questions whether complex social concepts such as fairness can be represented by mathematical definitions. A model can satisfy demographic parity or equal opportunity, but this does not automatically indicate that the whole system is fair. Hence, understanding algorithmic fairness requires looking at the environment in which it is being used (Selbst et al., 2019).

Conclusion

This review shows that algorithmic fairness is more complicated than simply making a model accurate or satisfying a single mathematical criterion. Different definitions of fairness can lead to different conclusions about the same model, meaning that we first need to decide what kind of equality we are trying to achieve. At the same time, bias may already exist in the data

before the model is even developed, and focusing only on the algorithm can cause us to ignore the wider environment in which it operates. For these reasons, algorithmic fairness should not be treated only as a technical problem. Mathematical measures can help us identify inequalities, but they cannot decide on their own what should be considered fair. Researchers and developers should therefore consider not only whether a model satisfies a fairness measure, but also whether that measure makes sense for the real-world situation in which the model is being used.

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